



A novel fuzzy neural network approach with triangular fuzzy information for the selection of logistics service providers

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Abstract

In this article, we presents a novel fuzzy neural network approach designed to address multi criteria decision making (MCDM) problems, specifically for selecting logistics service providers. The proposed decision making model integrates triangular fuzzy numbers (TFNs) with a triangular fuzzy Einstein weighted averaging (TFEWA) aggregation operator to enhance the decision making process under uncertainty. Initially, we discussed the concept of triangular fuzzy numbers, which allows for the representation of uncertain and imprecise data typically presented in real-world decision making environments. The operational laws, score function, and Hamming distance measures for TFNs are presented to ensure accurate handling of the fuzzy input data. The TFEWA aggregation operator, which is based on Einstein norms and plays a crucial role in aggregating expert opinions in the evaluation process. In the decision making process, we collect expert opinions regarding logistics service providers, expressed as TFNs, which are then processed through the fuzzy neural network model. After that, we apply the proposed decision making model to select the best logistics service providers. The TFEWA operator computes values at the hidden and output layers, and activation functions are applied to produce final output values. These outputs provide a ranked list of logistics service providers based on their overall performance across multiple criteria. The effectiveness of this novel approach is validated through a comparative analysis with existing MCDM methods. The results demonstrate that the triangular fuzzy neural network approach outperforms traditional methods in terms of flexibility, accuracy, and its ability to handle uncertain, fuzzy data. Our method provides a robust decision support system, capable of managing complex decision making tasks in logistics and other fields.

Keywords Fuzzy neural network · Triangular fuzzy neural network · Decision support model · Logistics service provider

1 Introduction

Third-party logistics (3PL) providers, also known as logistics service providers (LSPs) (Weijers et al. 2012; Lai 2004), specialize in the processing, storage, and delivery of products, including raw materials and finished goods. They offer cost-effective alternatives to building and managing warehousing capacity and multiple facilities. Which makes them ideal for large-scale e-commerce organizations, LSP services can range from simple shipping options to complex operations managing the client's entire supply chain. It is very important for companies considering LSPs to understand their logistical requirements to balance the benefits with the associated costs. There are many excellent logistics providers available, but it might be difficult to choose the best. In the 3PL supply chain model, a business contracts with a third-party service provider to handle its logistical needs. Three partners are involved in this model. Shipper/Manufacturer is the business that must control the supply chain is the one that manufactures or markets the goods. Logistics provider is a company that handles several logistical tasks for the seller, including distribution, inventory control, transportation, and storage, and occasionally even supply chain planning. Customer is the person or entity that buys or obtains the goods. Figure 1 illustrates the 3PL supply

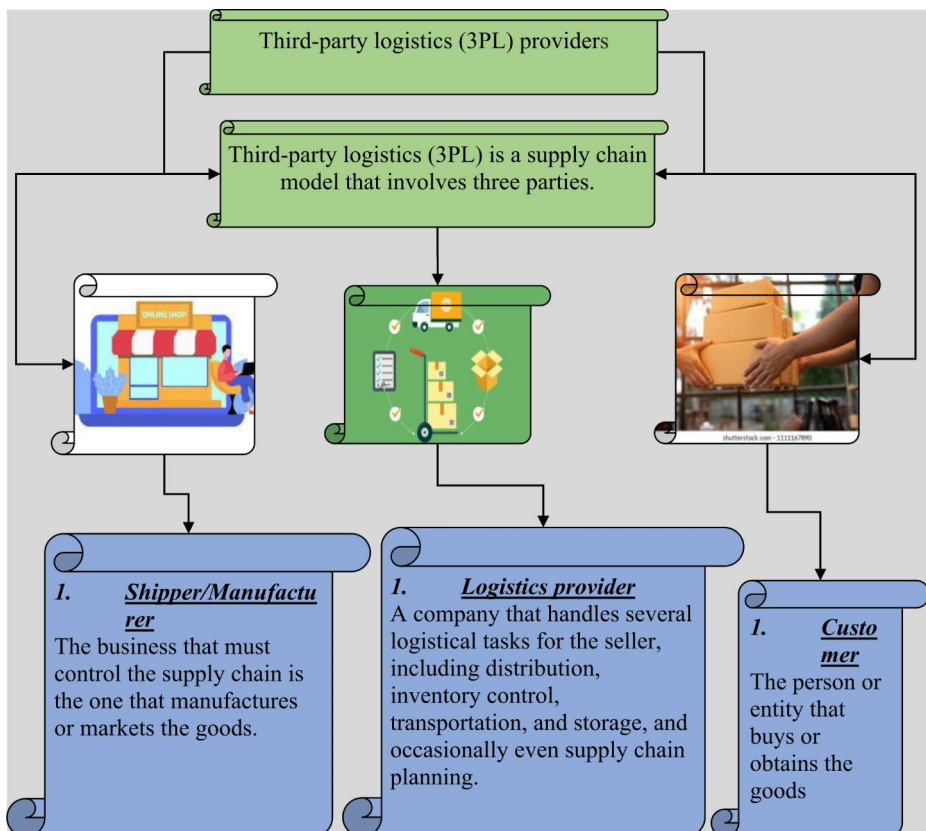


Fig. 1 The third-party logistics (3PL) supply chain model

chain model, providing a detailed representation of the processes and interactions involved in third-party logistics operations.

In recent years, the logistics industry has faced challenges in finding more efficient and flexible ways to handle various goods. Hybrid model for optimal selection of cold Chain logistics service providers (Krstić and Tadić 2023) is one solution to help choose the best cold chain service providers, which are important for transporting temperature-sensitive products. This model looks at both cost and service quality to make better decisions. Evaluating logistics flexibility in Istanbul-based companies using interval-valued Fermatean fuzzy SWARA (Korucuk and Aytekin 2023) helps companies in Istanbul adapt to changes in demand and unexpected issues, improving flexibility and ensuring smoother operations. With the rise of autonomous vehicles as an essential component of industry 4.0 for meeting last-mile logistics requirements (Dabic-Miletic 2023), these vehicles can improve delivery speed and reduce errors in last-mile logistics. The evaluating performance-based logistics in manufacturing through polytopic fuzzy SWARA (Aytekin and Korucuk 2024) helps companies assess logistics efficiency, focusing on cost, performance, and customer satisfaction to improve supply chain operations.

In modern decision making processes, multi criteria decision making (MCDM) (Bani-Doumi et al. 2024) is crucial. Real-world scenarios are often complicated, with uncertain criteria and subjective opinions influencing decisions. Decision makers (DMs) are tasked with providing evaluation data for different types of criteria, which can vary significantly. This complexity arises from the need to account for multiple factors and individual opinion. MCDM offers a structured approach to handle these challenges, enabling a comprehensive analysis and comparison of alternatives to make well-informed decisions in complex and uncertain environments.

The concept of fuzzy sets (FSs) has been a popular research topic since Zadeh introduced it in 1965 (Zadeh 1965). He also introduced fuzzy numbers (FNs) (Dubois and Prade 1993), which help represent uncertain and ambiguous information. However, working with FNs can be challenging because adding or multiplying them is not straightforward. Despite this, FNs are valuable because they represent human language-based information, making them useful in decision making. In 1986, Atanassov expanded on FSs by introducing intuitionistic fuzzy sets (IFs) (Atanassov and Stoeva 1986; Atanassov 1988). IFs add a non-membership degree, which provides more detailed and flexible information compared to FSs. Triangular fuzzy numbers (TFNs) (Anand and Bharatraj 2017; Gani and Assarudeen 2012) are particularly useful for managing uncertainty in decision making and solving problems. They are simpler to work with than interval numbers and have become widely adopted in various fields. TFNs are used to model imprecise data in applications such as risk assessment (Yang et al. 2013), performance evaluation (Wang et al. 2016), forecasting (Zeng et al. 2016), and matrix games (Li 2012). Their ability to represent fuzzy and uncertain information effectively makes them a valuable tool in situations where precise values are difficult to obtain. Later, Shu et al. (Shu et al. 2006) introduced triangular intuitionistic fuzzy numbers (TIFNs), a specific type of fuzzy number that enhances the representation of uncertainty by incorporating both membership and non-membership degrees. TIFNs are particularly useful because they provide more flexibility in modeling imprecise information, offering a better way to deal with uncertain or incomplete data compared to traditional fuzzy numbers. This makes TIFNs an important tool in areas such as decision-making and problem-solving, where precision is often hard to achieve.

Fuzzy Neural Networks (FNNs) (Nawaz et al. 2025; Zhang et al. 2025; Kruse 2008; Lin and Cunningham 1995) combine fuzzy logic and artificial neural networks (ANNs) to create a powerful model for handling complex, uncertain data. Fuzzy logic deals with ambiguous or unclear information by using fuzzy sets, while neural networks learn patterns from data and improve over time. This combination helps FNNs handle real-world problems where data is often imprecise or incomplete. In FNNs, a fuzzy rule base defines how inputs and outputs relate using fuzzy rules. These rules help the system process unclear information, using terms like “high” or “low” instead of exact numbers. This makes FNNs useful for decision making that involves subjective or qualitative data. The neural network adjusts the weights of connections between neurons to minimize errors, allowing the system to learn and improve its performance. The process begins with fuzzification, where exact data is converted into fuzzy values. This step is important when precise data isn’t available. After processing the fuzzy data and applying the rules, the system outputs fuzzy results, which are then turned into clear, actionable data through defuzzification.

FNNs are widely used in fields like MCDM, risk assessment, forecasting, and control systems, making them valuable tools for solving problems that require adaptability. They are capable of handling both unclear and exact data, which makes them highly effective in complex systems with uncertainty. For example, when predicting outcomes in complex systems, combining Fuzzy K-Clustering with FNNs (Wang et al. 2023) helps find patterns and make better predictions, especially when working with large amounts of messy data. By grouping data into fuzzy sets and using FNNs to learn from it, these systems can give more accurate forecasts. In power systems (Olusanya et al. 2024) like managing energy in factories, FNNs help improve energy efficiency and keep the power factor balanced. These systems can adjust to changing conditions by using fuzzy logic to deal with unclear input data like voltage changes, and neural networks fine-tune their actions over time. For transportation safety (Malik et al. 2023), FNNs combined with fuzzy logic help predict dangerous driving behaviors. By handling uncertain driving data, these systems can assess risks in real-time, leading to safer driving technologies. In floor segmentation for image processing, FNNs work with Fuzzy C-Means (FCM) and Convolutional Neural Networks (CNN) (Ravishankar et al. 2023) to make the system better at handling unclear or noisy images. FNNs help the system adapt to different lighting or angles, improving floor recognition accuracy.

Research background Real-world decision making problems are becoming more complex, importance the need for better decision making systems. The MCDM is a useful method to choose the best option from different alternatives, based on multiple factors. MCDM problems often involve balancing conflicting goals, such as cost, quality, and performance. Traditional MCDM methods such as TOPSIS (Abdullah et al. 2023a), GRA (Xie 2023), WASPAS (Radomska-Zalas 2023), MOORA (Chakraborty 2011), SWA (Kasim and Abdullah 2013), EDAS (Abdullah et al. 2023b) and CODAS (Shams and Abdullah 2023), have been widely used. However, these methods struggle when dealing with uncertainty, especially with unclear or subjective data. To overcome these challenges, researchers have combined fuzzy logic with MCDM methods. Fuzzy logic helps handle uncertainty by allowing partial membership instead of exact values, which is useful for dealing with ambiguous or imprecise data. Fuzzy Neural Networks, which combine fuzzy logic and neural networks, are a powerful solution. FNNs can process both fuzzy and numerical data, making them adaptable for decision making. Current FNN models have limitations, such as handling

imprecise data and integrating multiple factors. To solve this, this research proposes a new fuzzy neural network model using triangular fuzzy numbers, which provide a better way to represent uncertainty. This model also includes the Einstein aggregation operator, improving the integration of multiple factors, especially for complex MCDM problems like evaluating logistics service providers. This study aims to improve decision making models, especially in logistics and other important industries, helping decision makers make better choices and improve efficiency.

Motivation As decision-making problems in real life become more complex, there is an increasing need for better methods to solve them. In particular, MCDM is a field where decisions need to be made based on multiple factors. Researchers have explored different approaches to tackle these challenges using neural networks. For example, Abdullah et al. (Abdullah et al. 2024) used neural networks combined with the Hamacher aggregation operator to help choose the best network. Abosuliman et al. (Abosuliman et al. 2024) applied a fuzzy credibility neural network with the Dombi aggregation operator to select the best nanomaterials for nanosensors. Similarly, Zhang et al. (Zhang et al. 2024) proposed a method based on double hierarchy linguistic information to support robot selection. Building on these studies, our goal is to introduce a novel decision making model based on the triangular fuzzy neural network. This study aims to propose a new fuzzy neural network for MCDM problems to address the limitations of the traditional fuzzy logic and fuzzy neural network analyses. The main motivations behind this study are.

- i. To develop a novel approach for decision making based on a fuzzy neural network using triangular fuzzy numbers, which enhances the flexibility and accuracy of the decision making process.
- ii. To define the Einstein aggregation operator (Wang and Liu 2011; Fahmi et al. 2018; Iampan et al. 2021) for triangular fuzzy numbers, enabling more effective integration of multiple criteria in decision making.
- iii. To apply the proposed approach in analyzing logistics service providers, offering a practical solution for optimizing decision making in this critical industry.

Novelties This article introduces a novel approach for decision making based on fuzzy neural networks using triangular fuzzy information. We modify the standard feed-forward neural network to create a triangular fuzzy neural network for decision making problems. Additionally, we expand the Einstein aggregation operator to develop the triangular fuzzy Einstein aggregation operator. The proposed approach is applied to select the best logistics service provider. By considering the uncertainty and ambiguity in the performance data of logistics service providers, the model helps make more accurate and reliable decisions. This approach provides a flexible way to evaluate service providers by using triangular fuzzy values to represent uncertain or imprecise data. The model's framework can adjust to various decisions making situations, offering an effective solution for selecting logistics service providers.

Contribution In this article, we introduce a novel decision making model based on triangular fuzzy neural network designed to address complex decision problems with uncertain and imprecise data. The key contributions of this work are outlined below.

- i. **Develop a novel decision making model:** We develop a novel decision making model based on triangular fuzzy neural network to handle uncertain and fuzzy decision information. This model effectively processes imprecise data, offering more reliable and robust solutions to complex real-world decision making problems.
- ii. **Integrate the Einstein aggregation operator:** We integrate the Einstein aggregation operator into the neural network framework, enhancing the aggregation of multiple criteria. This integration improves the decision making process by capturing complex relationships between criteria, providing more accurate and flexible solutions for MCDM problems.
- iii. **Application to logistics service provider selection:** One of the main applications of our proposed model is in the selection of the best logistics service provider. We apply the TFNN approach to evaluate and rank various LSPs based on multiple criteria, including cost, quality, and performance. Our model takes into account the uncertainty and fuzziness in the performance data of different providers, offering decision-makers a more accurate and reliable way to choose the best LSP for their needs.
- iv. **Compare the proposed model with existing decision making models:** We compare the proposed model with several traditional decision making methods to validate its effectiveness. The results demonstrate that the proposed model outperforms these existing models, offering a more accurate and reliable solution for handling uncertainty and imprecision in decision making processes.

Structure of the article The main structure of the article is organized as follows.

- Section 2: This section discusses the TFN, its score function, and the Hamming distance measure for TFN. It also defines the operational laws for TFNs and the triangular fuzzy Einstein weighted averaging aggregation operator, and fuzzy neural networks.
- Section 3: This section explains the procedures for implementing a triangular fuzzy neural network in MCDM problems.
- Section 4: This section analyzes logistics service providers based on the proposed decision making model.
- Section 5: This section presents the results of the triangular fuzzy neural network and the selection of the best LSP.
- Section 6: This section compares the proposed decision making model with other MCDM models to evaluate its validity and efficiency.
- Section 7: This section outlines the advantages and limitations of the proposed model.
- Section 8: This section concludes the study and provides directions for future work.

Figure 2 presents the general structure of the article, offering an overview of its primary components and the logical progression of the content. It demonstrates the interconnections between key sections, such as the problem formulation, methodology, results, and conclusions. This diagram serves to clarify the overall framework and aids the reader in comprehending the organization of the article. Table 1 provides thorough explanations of the acronyms used in this article to help readers better comprehend the terms used. These acronyms are essential for simplifying technical language, and the table gives the reader a quick reference for better reading and comprehension.

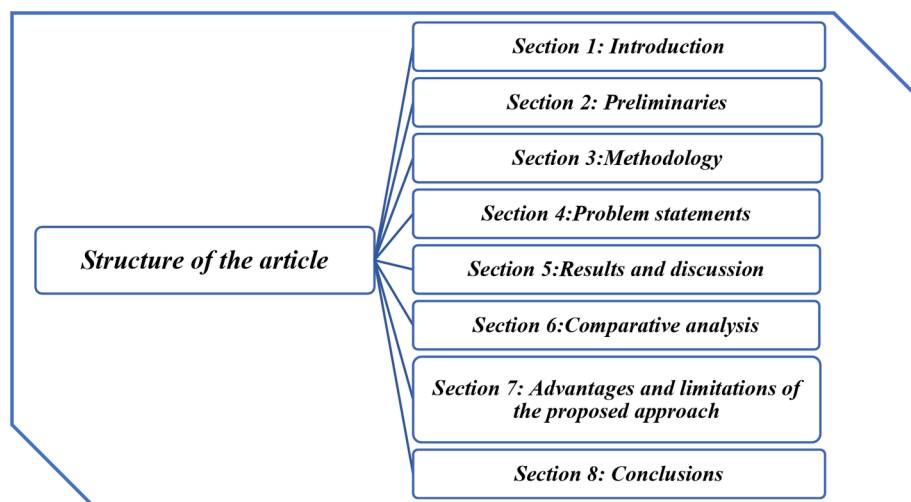


Fig. 2 The general structure of the article

Table 1 Description of the acronyms

Acronyms	Description
3PL	Third-party logistics
LSPs	Logistics service providers
MCDM	Multi criteria decision-making
DMM	Decision making model
DMs	Decision-makers
FSs	Fuzzy sets
FNs	Fuzzy numbers
IFSs	Intuitionistic fuzzy sets
TFNs	Triangular fuzzy numbers
FNNs	Fuzzy neural networks
LNNs	Linguistic neural network
FFNNs	Feed-forward neural networks
TFNN	Triangular fuzzy neural network
TFEWAAO	Triangular fuzzy Einstein weighted averaging aggregation operator
MD	Membership degree

Table 2 contains a complete list of all symbols and notations used in the present article, helping to ensure the mathematical expressions and discussion are clear and understandable.

2 Preliminaries

In this section, we discuss the TFN, its score function, and the Hamming distance measure for TFN. Based on Einstein norms, we define the operational laws for TFNs, the triangular fuzzy Einstein weighted averaging aggregation operator, and fuzzy neural networks.

Table 2 Description of the symbols

Symbols	Description
$\bar{\mathbb{U}}$	Universe set
$\hat{\mathcal{F}}$	Fuzzy set
$\mathcal{Q}_{\hat{\mathcal{F}}}(\sqcap)$	Membership degree
$\mathcal{T}$	Triangular fuzzy number
$\hat{\alpha}$	Lower bound of triangular fuzzy number
$\bar{\beta}$	Mode of triangular fuzzy number
$\bar{\gamma}$	Upper bound of triangular fuzzy number
$\hat{\mathcal{H}}(\mathcal{T})$	Score function of triangular fuzzy numbers
$\mathcal{D}(\mathcal{T}_1, \mathcal{T}_2)$	Distance measure between two triangular fuzzy numbers
$\mathcal{S}(\mathcal{T}_1, \mathcal{T}_2)$	Similarity measure between two triangular fuzzy numbers
$\mathbb{T}$	Einstein t-norm
$\mathbb{S}$	Einstein t-conorm
$\tilde{\omega}_i$	Expert Weights
η_r	Expert information
$\hat{\mathcal{M}}$	Alternatives
$\hat{\langle}$	Criteria
$\bar{\omega}_j$	Hidden layer weights
$\bar{\mathcal{T}}_{ij}$	Hidden layer information
$\bar{\mathcal{T}}_i$	Output layer information

Definition 1 Let $\bar{\mathbb{U}}$ be a universe set, and $\hat{\mathcal{F}}$ be a fuzzy set (Zadeh 1965) on $\bar{\mathbb{U}}$, defined as follows:

$$\hat{\mathcal{F}} = \left\{ (\sqcap, \mathcal{Q}_{\hat{\mathcal{F}}}(\sqcap)) \mid \sqcap \in \bar{\mathbb{U}} \right\}, \quad (1)$$

where the function $\mathcal{Q}_{\hat{\mathcal{F}}}(\sqcap): \bar{\mathbb{U}} \rightarrow [0,1]$ represents the membership degree.

Definition 2 A triangular fuzzy number (Anand and Bharatraj 2017; Gani and Assarudeen 2012) is a type of fuzzy number (Dubois and Prade 1993) that is defined by a triplet $\mathcal{T} = (\hat{\alpha}, \bar{\beta}, \bar{\gamma})$. The membership function of the triangular fuzzy number is defined as:

$$\mathcal{Q}_{\mathcal{T}}(\sqcap) = \begin{cases} 0, & \text{if } u < \hat{\alpha} \\ \frac{\sqcap - \hat{\alpha}}{\bar{\beta} - \hat{\alpha}}, & \text{if } \hat{\alpha} \leq u \leq \bar{\beta} \\ \frac{\bar{\gamma} - \sqcap}{\bar{\gamma} - \bar{\beta}}, & \text{if } \bar{\beta} \leq u \leq \bar{\gamma} \\ 0 & \text{if } u > \bar{\gamma} \end{cases} \quad (2)$$

The graphical representation of the triangular fuzzy number is shown in Fig. 3. This figure illustrates the structure and shape of a TFN. It provides a visual understanding of how the triangular fuzzy number is represented.

where $\hat{\alpha}$ is represents the lower bound, $\bar{\beta}$ represents is the mode, and $\check{\gamma}$ represents is the upper bound of the TFN. If $\hat{\alpha} \geq 0$ is called positive TFN and $\check{\gamma} \leq 0$ is called negative TFN. Figure 3 shows the graphical representation of TFN with $0 < \hat{\alpha} \leq \bar{\beta} \leq \check{\gamma}$. The membership function is classified by the following three points:

- i. Form $\hat{\alpha}$ to $\bar{\beta}$ is increasing function
- ii. From $\bar{\beta}$ to $\check{\gamma}$ is decreasing function

$$\hat{\alpha} \leq \bar{\beta} \leq \check{\gamma}$$

iii.

Definition 3 Let $\mathcal{T}_1 = (\hat{\alpha}_1, \bar{\beta}_1, \check{\gamma}_1)$ and $\mathcal{T}_2 = (\hat{\alpha}_2, \bar{\beta}_2, \check{\gamma}_2)$ be two TFNs and $\delta \geq 0$ then we have;

$$\mathcal{T}_1 + \mathcal{T}_2 = (\hat{\alpha}_1 + \hat{\alpha}_2, \bar{\beta}_1 + \bar{\beta}_2, \check{\gamma}_1 + \check{\gamma}_2)$$

i.

$$\mathcal{T}_1 \times \mathcal{T}_2 = (\hat{\alpha}_1 \times \hat{\alpha}_2, \bar{\beta}_1 \times \bar{\beta}_2, \check{\gamma}_1 \times \check{\gamma}_2)$$

ii.

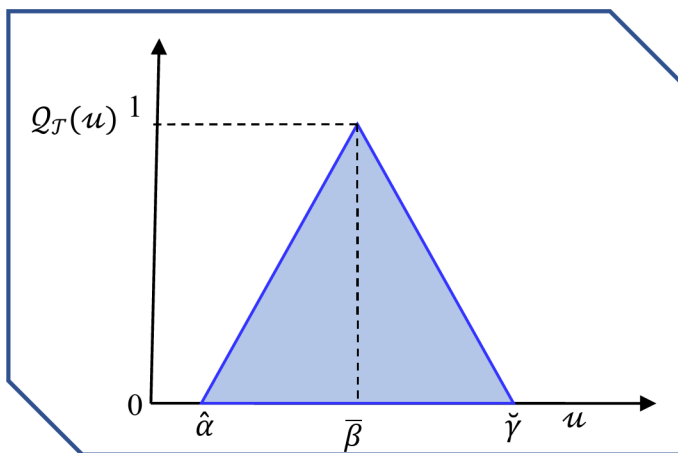


Fig. 3 The graphical representation of TFN

$$\delta \mathcal{T}_1 = \left(\delta \hat{\alpha}_1, \delta \bar{\beta}_1, \delta \check{\gamma}_1 \right)$$

iii.

$$\mathcal{T}_1^\delta = \left(\hat{\alpha}_1^\delta, \bar{\beta}_1^\delta, \check{\gamma}_1^\delta \right)$$

iv.

Definition 4 Let $\mathcal{T}_1 = \left(\hat{\alpha}_1, \bar{\beta}_1, \check{\gamma}_1 \right)$ and $\mathcal{T}_2 = \left(\hat{\alpha}_2, \bar{\beta}_2, \check{\gamma}_2 \right)$ be two TFNs. The Hamming distance between $\mathcal{T}_1$ and $\mathcal{T}_2$ defined as:

$$\mathcal{D}(\mathcal{T}_1, \mathcal{T}_2) = \sqrt{\frac{1}{3} \left\{ (\hat{\alpha}_2 - \hat{\alpha}_1)^2 + (\bar{\beta}_2 - \bar{\beta}_1)^2 + (\check{\gamma}_2 - \check{\gamma}_1)^2 \right\}} \quad (3)$$

Definition 5 Let $\mathcal{T}_1 = \left(\hat{\alpha}_1, \bar{\beta}_1, \check{\gamma}_1 \right)$ and $\mathcal{T}_2 = \left(\hat{\alpha}_2, \bar{\beta}_2, \check{\gamma}_2 \right)$ be two TFNs. Then the similarity measure between $\mathcal{T}_1$ and $\mathcal{T}_2$ defined as:

$$\mathcal{S}(\mathcal{T}_1, \mathcal{T}_2) = \frac{(\hat{\alpha}_1 + \hat{\alpha}_2) + (\bar{\beta}_1 + \bar{\beta}_2) + (\check{\gamma}_1 + \check{\gamma}_2)}{2 \left(\hat{\alpha}_1^2 + \bar{\beta}_1^2 + \check{\gamma}_1^2 \right) + 2 \left(\hat{\alpha}_2^2 + \bar{\beta}_2^2 + \check{\gamma}_2^2 \right) - \left((\hat{\alpha}_1 \times \hat{\alpha}_2) + (\bar{\beta}_1 \times \bar{\beta}_2) + (\check{\gamma}_1 \times \check{\gamma}_2) \right)} \quad (4)$$

Definition 6 Let $\mathbb{T}$ be Einstein t-norm and $\mathbb{S}$ be the t-conorm (Wang and Liu 2011) are shown as follows:

$$\mathbb{T}(x, y) = \frac{xy}{1 + (1 - x)(1 - y)} \quad (5)$$

$$\mathbb{S}(x, y) = \frac{x + y}{1 + xy} \quad (6)$$

Definition 7 Let $\mathcal{T}_1 = \left(\hat{\alpha}_1, \bar{\beta}_1, \check{\gamma}_1 \right)$ and $\mathcal{T}_2 = \left(\hat{\alpha}_2, \bar{\beta}_2, \check{\gamma}_2 \right)$ be two TFNs and $\delta > 0$. Then the operational Laws of TFNs based on Einstein norms defined as follow:

$$\mathcal{T}_1 \oplus_E \mathcal{T}_2 = \left(\frac{\hat{\alpha}_1 + \hat{\alpha}_2}{1 + \hat{\alpha}_1 \hat{\alpha}_2}, \frac{\bar{\beta}_1 + \bar{\beta}_2}{1 + \bar{\beta}_1 \bar{\beta}_2}, \frac{\check{\gamma}_1 + \check{\gamma}_2}{1 + \check{\gamma}_1 \check{\gamma}_2} \right)$$

i.

$$\mathcal{T}_1 \otimes_E \mathcal{T}_2 = \left(\frac{\widehat{\alpha}_1 \widehat{\alpha}_2}{1 + (1 - \widehat{\alpha}_1)(1 - \widehat{\alpha}_2)}, \frac{\bar{\beta}_1 \bar{\beta}_2}{1 + (1 - \bar{\beta}_1)(1 - \bar{\beta}_2)}, \frac{\check{\gamma}_1 \check{\gamma}_2}{1 + (1 - \check{\gamma}_1)(1 - \check{\gamma}_2)} \right)$$

ii.

$$\delta \odot_E \mathcal{T}_1 = \left(\frac{(1 + \widehat{\alpha}_1)^\delta - (1 - \widehat{\alpha}_1)^\delta}{(1 + \widehat{\alpha}_1)^\delta + (1 - \widehat{\alpha}_1)^\delta}, \frac{(1 + \bar{\beta}_1)^\delta - (1 - \bar{\beta}_1)^\delta}{(1 + \bar{\beta}_1)^\delta + (1 - \bar{\beta}_1)^\delta}, \frac{(1 + \check{\gamma}_1)^\delta - (1 - \check{\gamma}_1)^\delta}{(1 + \check{\gamma}_1)^\delta + (1 - \check{\gamma}_1)^\delta} \right)$$

iii.

$$\mathcal{T}_1^\delta = \left(\frac{2(\widehat{\alpha}_1)^\delta}{(2 - \widehat{\alpha}_1)^\delta + (\widehat{\alpha}_1)^\delta}, \frac{2(\bar{\beta}_1)^\delta}{(2 - \bar{\beta}_1)^\delta + (\bar{\beta}_1)^\delta}, \frac{2(\check{\gamma}_1)^\delta}{(2 - \check{\gamma}_1)^\delta + (\check{\gamma}_1)^\delta} \right)$$

iv.

Example 1 Let $\mathcal{T}_1 = (0.6, 0.8, 0.9)$ and $\mathcal{T}_2 = (0.5, 0.6, 0.8)$, be two triangular fuzzy numbers, with $\delta = 0.5$. Based on the defined operational laws, the results are as follows:

$$\mathcal{T}_1 \oplus_E \mathcal{T}_2 = (0.846, 0.946, 0.988)$$

i.

$$\mathcal{T}_1 \otimes_E \mathcal{T}_2 = (0.250, 0.444, 0.706)$$

ii.

$$\delta \odot_E \mathcal{T}_1 = (0.333, 0.500, 0.627)$$

iii.

$$\mathcal{T}_1^\delta = (0.791, 0.899, 0.950)$$

iv.

Theorem 1 Let $\mathcal{T}_1 = (\widehat{\alpha}_1, \bar{\beta}_1, \check{\gamma}_1)$ and $\mathcal{T}_2 = (\widehat{\alpha}_2, \bar{\beta}_2, \check{\gamma}_2)$ be two TFNs. Then we have;

$$\mathcal{T}_1 \oplus_E \mathcal{T}_2 = \mathcal{T}_2 \oplus_E \mathcal{T}_1;$$

i.

$$\mathcal{T}_1 \otimes_E \mathcal{T}_2 = \mathcal{T}_2 \otimes_E \mathcal{T}_1;$$

ii.

$$\delta (\mathcal{T}_1 \oplus_E \mathcal{T}_2) = \delta \mathcal{T}_1 \oplus_E \delta \mathcal{T}_2, \delta > 0;$$

iii.

$$(\delta_1 + \delta_2) \mathcal{T}_1 = \delta_1 \mathcal{T}_1 + \delta_2 \mathcal{T}_1, \delta_1 \text{ and } \delta_2 > 0;$$

iv.

$$(\mathcal{T}_1 \otimes_E \mathcal{T}_2)^\delta = \mathcal{T}_1^\delta \otimes_E \mathcal{T}_2^\delta, \delta > 0;$$

v.

$$\mathcal{T}_1^{\delta_1} \otimes_E \mathcal{T}_1^{\delta_2} = \mathcal{T}_1^{(\delta_1 + \delta_2)}, \delta_1 \text{ and } \delta_2 > 0$$

vi.

Definition 8 Let $\mathcal{T} = (\hat{\alpha}, \bar{\beta}, \check{\gamma})$ be a TFN, then the score function of TFN can be defined as:

$$\hat{\mathcal{H}}(\mathcal{T}) = \frac{\hat{\alpha} + \bar{\beta} + \check{\gamma}}{3}, \hat{\mathcal{H}}(\mathcal{T}) \in [0, 1], \quad (7)$$

Let $\hat{\mathcal{H}}(\mathcal{T})$ be the score function of TFN and $\mathcal{T}_1 = (\hat{\alpha}_1, \bar{\beta}_1, \check{\gamma}_1)$ and $\mathcal{T}_2 = (\hat{\alpha}_2, \bar{\beta}_2, \check{\gamma}_2)$ be two TFNs. Then we have;

- If $\hat{\mathcal{H}}(\mathcal{T}_1) < \hat{\mathcal{H}}(\mathcal{T}_2)$, then $\mathcal{T}_1 < \mathcal{T}_2$
- If $\hat{\mathcal{H}}(\mathcal{T}_1) > \hat{\mathcal{H}}(\mathcal{T}_2)$, then $\mathcal{T}_1 > \mathcal{T}_2$
- If $\hat{\mathcal{H}}(\mathcal{T}_1) = \hat{\mathcal{H}}(\mathcal{T}_2)$, then $\mathcal{T}_1 \sim \mathcal{T}_2$

Example 2 Let $\mathcal{T} = (0.6, 0.8, 0.9)$ be a triangular fuzzy number. The score value of $\mathcal{T}$ is 0.7670, calculated by using the score function (7) of TFNs.

Definition 9 Let $\mathcal{T}_i = (\hat{\alpha}_i, \bar{\beta}_i, \check{\gamma}_i) | i = 1, 2, 3, \dots, n$ be the collection of TFNs and

$\tilde{W} = (\tilde{\omega}_1, \tilde{\omega}_2, \tilde{\omega}_3, \dots, \tilde{\omega}_n)^T$ be the corresponding weight vector with $\tilde{\omega}_i \in [0, 1]$ and $\sum_{i=1}^n \tilde{\omega}_i = 1$. Then the triangular fuzzy Einstein weighted averaging (TFEWA) aggregation operator define as:

$$\begin{aligned} \ddot{\omega}_i(\mathcal{T}_1, \mathcal{T}_2, \mathcal{T}_3, \dots, \mathcal{T}_n) &= \oplus_{i=1}^n \ddot{\omega}_i \mathcal{T}_i \\ &= \left(\frac{\prod_{i=1}^n (1+\hat{\alpha}_i)^{\ddot{\omega}_i} - \prod_{i=1}^n (1-\hat{\alpha}_i)^{\ddot{\omega}_i}}{\prod_{i=1}^n (1+\hat{\alpha}_i)^{\ddot{\omega}_i} + \prod_{i=1}^n (1-\hat{\alpha}_i)^{\ddot{\omega}_i}}, \frac{\prod_{i=1}^n (1+\bar{\beta}_i)^{\ddot{\omega}_i} - \prod_{i=1}^n (1-\bar{\beta}_i)^{\ddot{\omega}_i}}{\prod_{i=1}^n (1+\bar{\beta}_i)^{\ddot{\omega}_i} + \prod_{i=1}^n (1-\bar{\beta}_i)^{\ddot{\omega}_i}}, \right. \\ &\quad \left. \frac{\prod_{i=1}^n (1+\check{\gamma}_i)^{\ddot{\omega}_i} - \prod_{i=1}^n (1-\check{\gamma}_i)^{\ddot{\omega}_i}}{\prod_{i=1}^n (1+\check{\gamma}_i)^{\ddot{\omega}_i} + \prod_{i=1}^n (1-\check{\gamma}_i)^{\ddot{\omega}_i}} \right) \quad (8) \end{aligned}$$

Example 3 Let $\{(0.6, 0.7, 0.8), (0.7, 0.8, 0.9), (0.5, 0.6, 0.7)\}$ represent a collection of triangular fuzzy numbers and $(0.33, 0.32, 0.35)$ be the corresponding weight vector. These values are aggregated using the triangular fuzzy Einstein weighted averaging aggregation operator and the results are as follows:

$$\begin{aligned} \ddot{\omega}_i(\mathcal{T}_1, \mathcal{T}_2, \mathcal{T}_3) &= \oplus_{i=1}^3 \ddot{\omega}_i \mathcal{T}_i = \left(\frac{\prod_{i=1}^3 (1+\hat{\alpha}_i)^{\ddot{\omega}_i} - \prod_{i=1}^3 (1-\hat{\alpha}_i)^{\ddot{\omega}_i}}{\prod_{i=1}^3 (1+\hat{\alpha}_i)^{\ddot{\omega}_i} + \prod_{i=1}^3 (1-\hat{\alpha}_i)^{\ddot{\omega}_i}}, \frac{\prod_{i=1}^3 (1+\bar{\beta}_i)^{\ddot{\omega}_i} - \prod_{i=1}^3 (1-\bar{\beta}_i)^{\ddot{\omega}_i}}{\prod_{i=1}^3 (1+\bar{\beta}_i)^{\ddot{\omega}_i} + \prod_{i=1}^3 (1-\bar{\beta}_i)^{\ddot{\omega}_i}}, \right. \\ &\quad \left. \frac{\prod_{i=1}^3 (1+\check{\gamma}_i)^{\ddot{\omega}_i} - \prod_{i=1}^3 (1-\check{\gamma}_i)^{\ddot{\omega}_i}}{\prod_{i=1}^3 (1+\check{\gamma}_i)^{\ddot{\omega}_i} + \prod_{i=1}^3 (1-\check{\gamma}_i)^{\ddot{\omega}_i}} \right) \\ &= \left(\frac{\begin{aligned} &<(1+0.6)^{0.33} \times (1+0.7)^{0.32} \times (1+0.5)^{0.35}> - <(1-0.6)^{0.33} \times (1-0.7)^{0.32} \times (1-0.5)^{0.35}> \\ &<(1+0.6)^{0.33} \times (1+0.7)^{0.32} \times (1+0.5)^{0.35}> + <(1-0.6)^{0.33} \times (1-0.7)^{0.32} \times (1-0.5)^{0.35}>, \\ &<(1+0.7)^{0.33} \times (1+0.8)^{0.32} \times (1+0.6)^{0.35}> - <(1-0.7)^{0.33} \times (1-0.8)^{0.32} \times (1-0.6)^{0.35}> \\ &<(1+0.7)^{0.33} \times (1+0.8)^{0.32} \times (1+0.6)^{0.35}> + <(1-0.7)^{0.33} \times (1-0.8)^{0.32} \times (1-0.6)^{0.35}>, \\ &<(1+0.8)^{0.33} \times (1+0.9)^{0.32} \times (1+0.7)^{0.35}> - <(1-0.8)^{0.33} \times (1-0.9)^{0.32} \times (1-0.7)^{0.35}> \\ &<(1+0.8)^{0.33} \times (1+0.9)^{0.32} \times (1+0.7)^{0.35}> + <(1-0.8)^{0.33} \times (1-0.9)^{0.32} \times (1-0.7)^{0.35}> \end{aligned}}{\begin{aligned} &<(1+0.6)^{0.33} \times (1+0.7)^{0.32} \times (1+0.5)^{0.35}> - <(1-0.6)^{0.33} \times (1-0.7)^{0.32} \times (1-0.5)^{0.35}> \\ &<(1+0.6)^{0.33} \times (1+0.7)^{0.32} \times (1+0.5)^{0.35}> + <(1-0.6)^{0.33} \times (1-0.7)^{0.32} \times (1-0.5)^{0.35}>, \\ &<(1+0.7)^{0.33} \times (1+0.8)^{0.32} \times (1+0.6)^{0.35}> - <(1-0.7)^{0.33} \times (1-0.8)^{0.32} \times (1-0.6)^{0.35}> \\ &<(1+0.7)^{0.33} \times (1+0.8)^{0.32} \times (1+0.6)^{0.35}> + <(1-0.7)^{0.33} \times (1-0.8)^{0.32} \times (1-0.6)^{0.35}>, \\ &<(1+0.8)^{0.33} \times (1+0.9)^{0.32} \times (1+0.7)^{0.35}> - <(1-0.8)^{0.33} \times (1-0.9)^{0.32} \times (1-0.7)^{0.35}> \\ &<(1+0.8)^{0.33} \times (1+0.9)^{0.32} \times (1+0.7)^{0.35}> + <(1-0.8)^{0.33} \times (1-0.9)^{0.32} \times (1-0.7)^{0.35}> \end{aligned}} \right) \\ &= \left(\frac{1.595 - 0.394}{1.595 + 0.394}, \frac{1.695 - 0.291}{1.695 + 0.291}, \frac{1.795 - 0.185}{1.795 + 0.185} \right) = (0.603, 0.707, 0.813) \end{aligned}$$

Definition 10 Fuzzy Neural Networks (FNN) (Nawaz et al. 2025; Zhang et al. 2025; Kruse 2008; Lin and Cunningham 1995) integrate fuzzy logic with artificial neural networks to handle uncertainty and imprecision in data. In this framework, input signals are represented as fuzzy numbers, which allows the network to work with vague or uncertain data. The fuzzy rules are used to map the fuzzy inputs to output values, enabling the system to make decisions based on incomplete or ambiguous information. The fuzzy neural network operates by processing input signals denoted as $\overline{\mathfrak{T}}_i = \left(\check{\mathfrak{T}}_1, \check{\mathfrak{T}}_2, \check{\mathfrak{T}}_3, \dots, \check{\mathfrak{T}}_n \right)$, where each input is a fuzzy number. Along with these fuzzy inputs, corresponding weights, denoted as $\overline{\mathfrak{w}} = \left(\check{\mathfrak{w}}_1, \check{\mathfrak{w}}_2, \check{\mathfrak{w}}_3, \dots, \check{\mathfrak{w}}_n \right)$, are assigned to each input signal. These weights reflect the relative importance or contribution of each input signal to the network's decision making process. The combination of fuzzy inputs and their corresponding weights is crucial to the operation of the fuzzy neural network. Each input signal is multiplied by its corresponding weight, indicating the strength or significance of that input in the decision making process. These weighted inputs are then processed through the network, using fuzzy rules and activation functions to generate the output. The process allows the network to handle complex, MCDM problems, making it suitable for applications that require consideration of imprecise or uncertain data. Figure 4 shows a graphical representation of the fuzzy neural

network, outlining the interaction between inputs, weights, hidden layers, and output in the decision making process.

3 Methodology

In this section, we discuss the procedures of triangular fuzzy neural network for the MCDM problems.

A triangular fuzzy neural network (TFNN) is based on the integration of two essential frameworks: fuzzy neural networks and triangular fuzzy information. FFNNs add fuzzy logic approaches to standard neural networks. Fuzzy logic is excellent for coping with uncertainty and ambiguity, allowing FNNs to properly interpret imprecise or ambiguous data. This makes them ideal for instances in which the data is not binary but has variable degrees of truth. TFNN are an effective tool for handling multi criteria, real-world situations including qualitative judgments and uncertainties because they combine these two ideas to manage both imprecise numerical data and complicated verbal input. Consequently, it is first necessary to define the TFNs information systems. Let $\eta_r = \{\eta_1, \eta_2, \dots, \eta_l\}$ be the l expert information. Suppose $\widehat{\mathcal{M}} = \{\widehat{\mathcal{M}}_1, \widehat{\mathcal{M}}_2, \dots, \widehat{\mathcal{M}}_m\}$ are a different set of alternatives, and $\widehat{\gamma} = \{\widehat{\gamma}_1, \widehat{\gamma}_2, \dots, \widehat{\gamma}_n\}$ are finite set of n criteria. The triangular fuzzy neural network (TFNN) consists of the following steps.

Step 1 The collection of expert opinion information matrices in the form of the TFNs $\tau_{ij} = (\hat{\alpha}_{ij}, \bar{\beta}_{ij}, \tilde{\gamma}_{ij})$. The expert opinion information is considering the input signals of the TFNN.

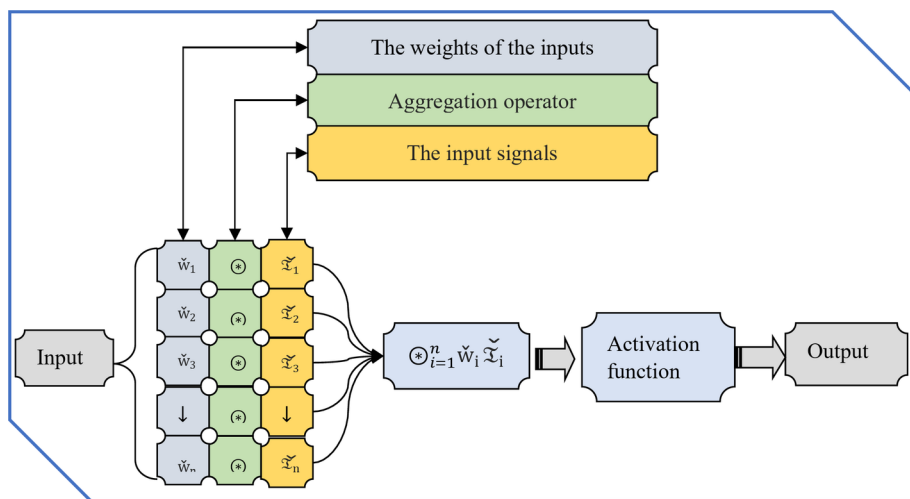


Fig. 4 The fuzzy neural network

$$\eta_r = \begin{bmatrix} \begin{pmatrix} \hat{\alpha}_{11}, \bar{\beta}_{11}, \check{\gamma}_{11} \\ \hat{\alpha}_{21}, \bar{\beta}_{21}, \check{\gamma}_{21} \\ \hat{\alpha}_{31}, \bar{\beta}_{31}, \check{\gamma}_{31} \\ \vdots \\ \hat{\alpha}_{m1}, \bar{\beta}_{m1}, \check{\gamma}_{m1} \end{pmatrix} & \begin{pmatrix} \hat{\alpha}_{12}, \bar{\beta}_{12}, \check{\gamma}_{12} \\ \hat{\alpha}_{22}, \bar{\beta}_{22}, \check{\gamma}_{22} \\ \hat{\alpha}_{32}, \bar{\beta}_{32}, \check{\gamma}_{32} \\ \vdots \\ \hat{\alpha}_{m2}, \bar{\beta}_{m2}, \check{\gamma}_{m2} \end{pmatrix} & \cdots & \begin{pmatrix} \hat{\alpha}_{1n}, \bar{\beta}_{1n}, \check{\gamma}_{1n} \\ \hat{\alpha}_{2n}, \bar{\beta}_{2n}, \check{\gamma}_{2n} \\ \hat{\alpha}_{3n}, \bar{\beta}_{3n}, \check{\gamma}_{3n} \\ \vdots \\ \hat{\alpha}_{mn}, \bar{\beta}_{mn}, \check{\gamma}_{mn} \end{pmatrix} \end{bmatrix} \quad (9)$$

Step 2 Calculate the expert weights based on the expert opinion information matrices. Let $\bar{\omega}_j$ be the expert weights based on the expert opinion information matrices such that $\bar{\omega}_j > 0$ and $\sum_{j=1}^n \bar{\omega}_j = 1$.

Step 3 Calculate the hidden layer information of the TFNN by using the Einstein aggregation operator to aggregate the input signals with the corresponding weights.

$$\bar{\tau}_{ij} = \oplus_{r=1}^l \bar{\omega}_j (\tau_{ij\eta_r}) = \left(\frac{\prod_{r=1}^l (1 + \hat{\alpha}_{ij})^{\bar{\omega}_j} - \prod_{r=1}^l (1 - \hat{\alpha}_{ij})^{\bar{\omega}_j}}{\prod_{r=1}^l (1 + \hat{\alpha}_{ij})^{\bar{\omega}_j} + \prod_{r=1}^l (1 - \hat{\alpha}_{ij})^{\bar{\omega}_j}}, \frac{\prod_{r=1}^l (1 + \bar{\beta}_{ij})^{\bar{\omega}_j} - \prod_{r=1}^l (1 - \bar{\beta}_{ij})^{\bar{\omega}_j}}{\prod_{r=1}^l (1 + \bar{\beta}_{ij})^{\bar{\omega}_j} + \prod_{r=1}^l (1 - \bar{\beta}_{ij})^{\bar{\omega}_j}}, \frac{\prod_{r=1}^l (1 + \check{\gamma}_{ij})^{\bar{\omega}_j} - \prod_{r=1}^l (1 - \check{\gamma}_{ij})^{\bar{\omega}_j}}{\prod_{r=1}^l (1 + \check{\gamma}_{ij})^{\bar{\omega}_j} + \prod_{r=1}^l (1 - \check{\gamma}_{ij})^{\bar{\omega}_j}} \right), \quad (10)$$

$$\text{where } \bar{\tau}_{ij} = \left(\bar{\hat{\alpha}}_{ij}, \bar{\beta}_{ij}, \bar{\check{\gamma}}_{ij} \right)$$

Step 4 Find the criteria weights of the hidden layer information. Let $\bar{\bar{\omega}}_j$ be the criteria weights of hidden layer information such that $\bar{\bar{\omega}}_j > 0$ and $\sum_{j=1}^n \bar{\bar{\omega}}_j = 1$.

Step 5 Calculate the output layer information by using the Einstein aggregation operator to aggregate the hidden layer information with the corresponding weights.

$$\bar{\bar{\tau}}_i = \oplus_{j=1}^n \bar{\bar{\omega}}_j \bar{\tau}_{ij} = \left(\frac{\prod_{j=1}^n (1 + \bar{\hat{\alpha}}_{ij})^{\bar{\bar{\omega}}_j} - \prod_{j=1}^n (1 - \bar{\hat{\alpha}}_{ij})^{\bar{\bar{\omega}}_j}}{\prod_{j=1}^n (1 + \bar{\hat{\alpha}}_{ij})^{\bar{\bar{\omega}}_j} + \prod_{j=1}^n (1 - \bar{\hat{\alpha}}_{ij})^{\bar{\bar{\omega}}_j}}, \frac{\prod_{j=1}^n (1 + \bar{\beta}_{ij})^{\bar{\bar{\omega}}_j} - \prod_{j=1}^n (1 - \bar{\beta}_{ij})^{\bar{\bar{\omega}}_j}}{\prod_{j=1}^n (1 + \bar{\beta}_{ij})^{\bar{\bar{\omega}}_j} + \prod_{j=1}^n (1 - \bar{\beta}_{ij})^{\bar{\bar{\omega}}_j}}, \frac{\prod_{j=1}^n (1 + \bar{\check{\gamma}}_{ij})^{\bar{\bar{\omega}}_j} - \prod_{j=1}^n (1 - \bar{\check{\gamma}}_{ij})^{\bar{\bar{\omega}}_j}}{\prod_{j=1}^n (1 + \bar{\check{\gamma}}_{ij})^{\bar{\bar{\omega}}_j} + \prod_{j=1}^n (1 - \bar{\check{\gamma}}_{ij})^{\bar{\bar{\omega}}_j}} \right), \quad (11)$$

$$\text{where } \bar{\bar{\tau}}_i = \left(\bar{\bar{\hat{\alpha}}}_i, \bar{\bar{\beta}}_i, \bar{\bar{\check{\gamma}}}_i \right).$$

Step 6 Calculate the score values of output layer information by using the score functions of TFN.

$$\hat{\mathcal{H}}\left(\overline{\overline{\overline{\mathcal{T}}}}_i\right) = \frac{\overline{\overline{\overline{\hat{\alpha}}}}_i + \overline{\overline{\overline{\hat{\beta}}}}_i + \overline{\overline{\overline{\hat{\gamma}}}}_i}{3}, \hat{\mathcal{H}}\left(\overline{\overline{\overline{\mathcal{T}}}}_i\right) \in [0,1] \quad (12)$$

Step 7 To calculate the output information of TFNN by using the sigmoid activation function (Pratiwi et al. 2020), Softmax activation function (Raghuram et al. 2022), hyperbolic tangent (Tanh) activation function (Kumar and Sodhi 2022), and Swish activation function (Avenash and Viswanath 2019). The activation function in the output layer of TFNN is essential to transform the raw output values into a meaningful range. The choice of activation function depends on the task at hand. The Sigmoid function is typically used for binary classification problems, where the output is between 0 and 1, representing a probability. The Softmax function is ideal for multi-class classification, as it normalizes the output into a probability distribution where the sum of all outputs equals 1. The Tanh function is useful when the outputs need to be in the range of -1 to 1 , often used in regression or tasks requiring zero-centered outputs. The Swish function is a newer activation function that allows for smoother gradients and can improve performance in deep learning tasks, especially when capturing complex patterns.

$$\text{Sigmoid Activation Function} = \mathcal{F}_1\left(\hat{\mathcal{H}}\left(\overline{\overline{\overline{\mathcal{T}}}}_i\right)\right) = \frac{1}{1 + e^{-\left(\hat{\mathcal{H}}\left(\overline{\overline{\overline{\mathcal{T}}}}_i\right)\right)}} \quad (13)$$

$$\text{Softmax Activation Function} = \mathcal{F}_2\left(\hat{\mathcal{H}}\left(\overline{\overline{\overline{\mathcal{T}}}}_i\right)\right) = \frac{e^{-\left(\hat{\mathcal{H}}\left(\overline{\overline{\overline{\mathcal{T}}}}_i\right)\right)}}{\sum_{i=1}^m e^{-\left(\hat{\mathcal{H}}\left(\overline{\overline{\overline{\mathcal{T}}}}_i\right)\right)}} \quad (14)$$

$$\text{Hyperbolic Tangent Activation Function} = \mathcal{F}_3\left(\hat{\mathcal{H}}\left(\overline{\overline{\overline{\mathcal{T}}}}_i\right)\right) = \frac{e^{\left(\hat{\mathcal{H}}\left(\overline{\overline{\overline{\mathcal{T}}}}_i\right)\right)} - e^{-\left(\hat{\mathcal{H}}\left(\overline{\overline{\overline{\mathcal{T}}}}_i\right)\right)}}{e^{\left(\hat{\mathcal{H}}\left(\overline{\overline{\overline{\mathcal{T}}}}_i\right)\right)} + e^{-\left(\hat{\mathcal{H}}\left(\overline{\overline{\overline{\mathcal{T}}}}_i\right)\right)}} \quad (15)$$

$$\text{Swish Activation Function} = \mathcal{F}_4 \left(\hat{\mathcal{H}} \left(\frac{-}{\mathcal{T}_i} \right) \right) = \frac{\hat{\mathcal{H}} \left(\frac{-}{\mathcal{T}_i} \right)}{1 + e^{-\left(\hat{\mathcal{H}} \left(\frac{-}{\mathcal{T}_i} \right) \right)}} \quad (16)$$

Figure 5 illustrates the detailed procedures of the triangular fuzzy neural network used for solving MCDM problems. The figure outlines the process of the TFNN model to evaluate various alternatives based on multiple decisions making criteria, ultimately identifying the optimal choice.

4 Problem statements

Third-party logistics (3PL) providers, also known as logistics service providers (LSPs) (Weijers et al. 2012; Lai 2004), are experts in handling the storage, processing, and delivery of products, which can include both raw materials and finished goods. These providers offer businesses a cost-effective solution, eliminating the need to invest in building and maintaining their own warehouses and facilities. This is especially beneficial for large e-commerce companies that require efficient logistics support. LSP services vary widely, from simple shipping solutions to more advanced services where the provider manages the entire supply chain for the client. For businesses, it is important to carefully assess their logistical needs, weighing the benefits against the costs involved in working with an LSP. The selection of the right provider is critical, as it can directly impact the company's operational efficiency and overall success. Given the wide range of high-quality providers in the market, businesses often find it difficult to choose the most suitable one for their needs. In this paper, we applied the triangular fuzzy neural network model to help in the selection process. The TFNN model uses fuzzy logic to handle the uncertainties that come with decision making, allowing businesses to evaluate logistics service providers based on multiple criteria such as cost, reliability, and service quality. This approach allows companies to make a more informed decision, ensuring that they select the LSP that will best meet their needs and contribute to efficient supply chain management. We used this model to evaluate five different LSPs, helping to identify the best provider for the business's requirements.

Total quality logistics-($\widehat{\mathcal{M}}_1$) Total Quality Logistics (TQL) (Thai and Jie 2018) is a well-known logistics company in the United States. It offers services like freight brokerage, transportation management, and supply chain solutions. TQL helps businesses transport goods efficiently and at a lower cost. They handle different types of shipments, such as truckload and less-than-truckload, both within the U.S. and internationally. TQL uses technology to track shipments and keep customers informed in real-time. The company is known for its strong customer service and focus on providing reliable, cost-effective solutions for businesses, making it a trusted choice in the logistics industry.

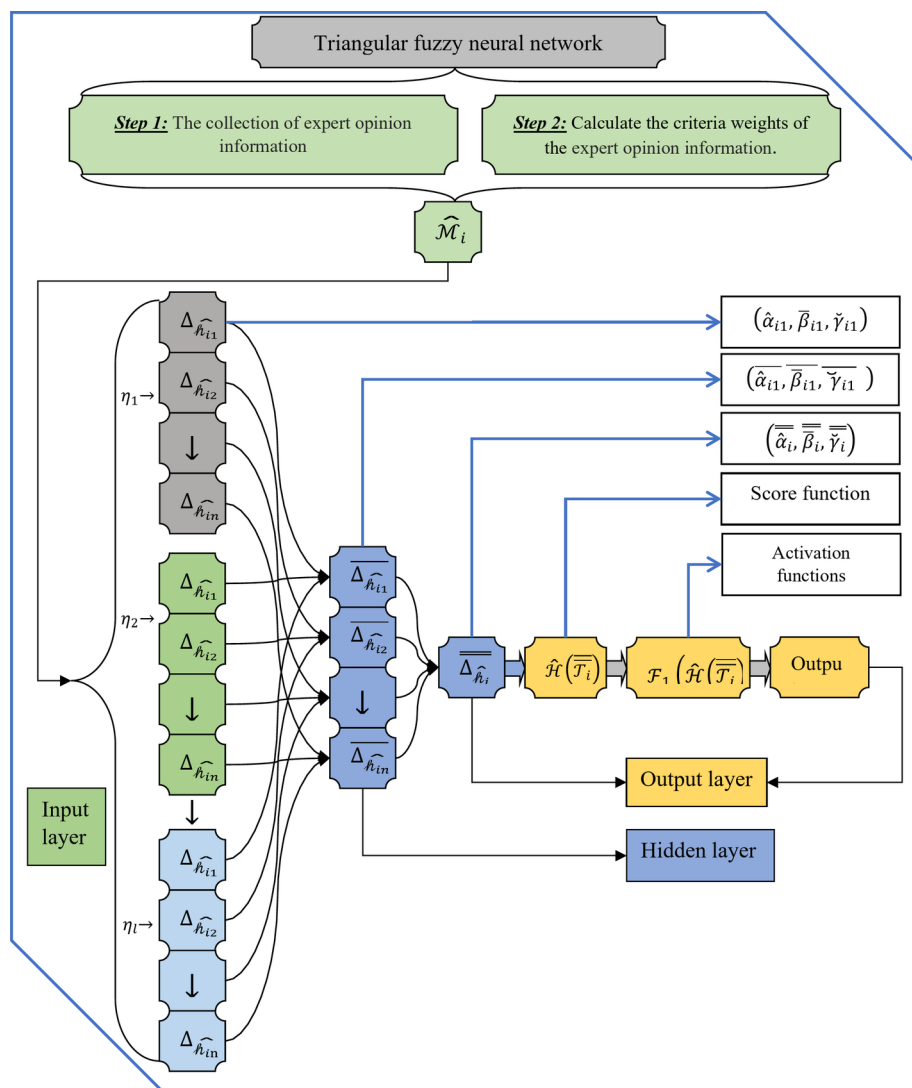


Fig. 5 Detailed process flow of the triangular fuzzy neural network

GXO logistics-($\hat{\mathcal{M}}_2$) GXO Logistics (Grover and Ashraf 2024) is a global logistics company that provides supply chain solutions to businesses. It offers services such as warehousing, transportation, and inventory management. GXO works with companies to improve their distribution processes by using advanced technology and automation to increase efficiency. The company serves a variety of industries, including retail, e-commerce, healthcare, and automotive. GXO's services are designed to help businesses lower costs, speed up deliveries, and enhance customer satisfaction. With a strong focus on innovation and sustainability, GXO is a key player in the logistics and supply chain sector.

UPS supply chain solutions-($\widehat{\mathcal{M}}_3$) UPS Supply Chain Solutions (Jaboob et al. 2024) is a division of UPS that offers comprehensive logistics and supply chain services to businesses worldwide. The company provides a range of services, including warehousing, distribution, transportation management, and customs brokerage. UPS Supply Chain Solutions helps businesses streamline their operations by offering integrated solutions that improve efficiency, reduce costs, and enhance customer satisfaction. With its global network and advanced technology, UPS enables businesses to manage inventory, track shipments, and optimize supply chains. The company serves various industries, such as retail, healthcare, automotive, and technology, offering tailored solutions to meet specific needs.

Expeditors international of Washington-($\widehat{\mathcal{M}}_4$) Expeditors International of Washington (Min and Joo 2009) is a global logistics and supply chain Management Company that provides transportation, warehousing, and distribution services. They specialize in managing the movement of goods across various modes of transport, including air, ocean, and ground. Expeditors' offers services such as customs brokerage, freight forwarding, and inventory management, helping businesses streamline their supply chains. With a strong focus on customer service and technology, Expeditors ensures the efficient movement of goods and accurate tracking. The company operates worldwide, serving industries like retail, manufacturing, automotive, and healthcare.

C.H. Robinson Worldwide-($\widehat{\mathcal{M}}_5$) C.H. Robinson Worldwide (Host 2022) is a global logistics company that provides a range of services, including freight transportation, supply chain management, and sourcing solutions. They offer services like truckload, less-than-truckload shipping, air and ocean freight, and customs brokerage. C.H. Robinson helps businesses manage their supply chains more efficiently by connecting them with carriers and suppliers worldwide. Their technology platform provides real-time tracking and data analytics to optimize logistics operations. With a focus on customer service and innovation, C.H. Robinson serves various industries, including retail, automotive, food, and manufacturing.

Figure 6 illustrates the steps involved in selecting the best Logistics Service Provider. It outlines the procedure followed to evaluate and rank different LSPs. This visual guide helps in understanding the selection process clearly.

The selection of Logistics Service Providers (LSPs) is based on four key criteria, which we have considered for evaluation:

- **Compatibility-($\widehat{1}$):** The compatibility (Omoruyi 2020) between the user and the logistics provider, Emphasizes their mutual efforts to achieve common goals. Characteristics related to business plan effectiveness, technical intelligence, corporate culture, and other logistics service providers all fall under the umbrella of competencies.
- **Delivery performance-($\widehat{2}$):** Incorporating elements such as “speed,” “safety,” and “security,” is important for ensuring customer satisfaction and achieving efficient and effective logistics performance (Domingues et al. 2015) with high cost.
- **Management-($\widehat{3}$):** the management's (Forslund 2012) expertise in delivering exceptional services while maintaining a competitive edge in the market.
- **Cost-($\widehat{4}$):** Referring to the overall costs (Jiang et al. 2016) of logistics outsourcing, this standard covers the management of all logistics activities within defined budgetary

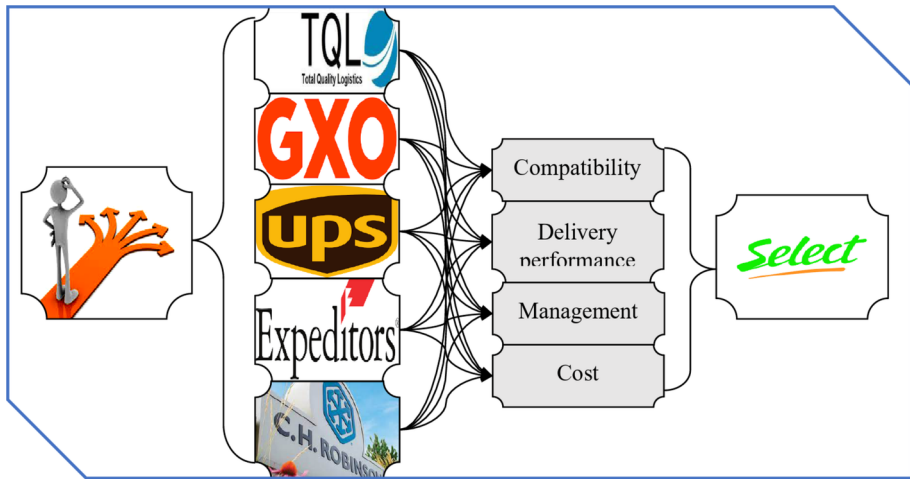


Fig. 6 The procedures for the selection of the best LSP

parameters.

5 Results and discussion

In this section, we discuss the results of the triangular fuzzy neural network and the selection of the best LSP.

Step 1 Three specialists provided opinions about the logistics service providers, and we collected their evaluations into information matrices. TFNs are used to indicate each expert's assessment. By providing each expert's view as a triplet of values a lower limit, a most probable value, and an upper limit these TFNs aid in the assessment of uncertainty and opinions that are subjective. Their judgments of the performance of the logistics service providers are rendered more flexible and thorough by this method, which enables us to identify and examine the inherent ambiguity in their evaluations.

Step 2 We use known weights based on expert opinion information matrices. These weights are crucial to the decision-making process because they represent the weight assigned to each expert based on their professional judgments, which will be used in subsequent calculations or analyses that form the framework for decision making.

Step 3 In TFNN, we compute the hidden layer information by using the Einstein aggregation operator (10). This operator makes sure that the fuzziness present in input signals is appropriately taken into account by combining the signals with their weights. Each input signal in the hidden layer goes through the same procedure, and Table 3 displays the hidden layer information. The information of the hidden layer is essential to the continued transmission of signals across the network. The information of the hidden layer is more dependable and precise thanks to this non-linear mixing of input signals and weights.

Table 3 The hidden layer information of the TFNN

LSPs	Compatibility	Delivery performance
$\widehat{\mathcal{M}}_1$	(0.428, 0.562, 0.662)	(0.377, 0.520, 0.606)
$\widehat{\mathcal{M}}_2$	(0.592, 0.732, 0.832)	(0.382, 0.512, 0.578)
$\widehat{\mathcal{M}}_3$	(0.494, 0.646, 0.740)	(0.513, 0.684, 0.714)
$\widehat{\mathcal{M}}_4$	(0.470, 0.542, 0.704)	(0.519, 0.637, 0.768)
$\widehat{\mathcal{M}}_5$	(0.625, 0.756, 0.821)	(0.416, 0.529, 0.607)
LSPs	Management	Cost
$\widehat{\mathcal{M}}_1$	(0.562, 0.671, 0.729)	(0.437, 0.578, 0.732)
$\widehat{\mathcal{M}}_2$	(0.372, 0.475, 0.593)	(0.502, 0.612, 0.692)
$\widehat{\mathcal{M}}_3$	(0.421, 0.531, 0.677)	(0.356, 0.485, 0.717)
$\widehat{\mathcal{M}}_4$	(0.585, 0.690, 0.801)	(0.515, 0.694, 0.805)
$\widehat{\mathcal{M}}_5$	(0.662, 0.758, 0.805)	(0.662, 0.776, 0.805)

Table 4 The output layer information of the TFNN

LSPs	$\overline{\overline{\mathcal{T}}} = \left(\overline{\overline{\alpha}}, \overline{\overline{\beta}}, \overline{\overline{\gamma}} \right)$	Score values
$\mathcal{M}_1$	(0.464, 0.593, 0.688)	0.5817
$\mathcal{M}_2$	(0.464, 0.589, 0.689)	0.5807
$\mathcal{M}_3$	(0.448, 0.590, 0.711)	0.5833
$\mathcal{M}_4$	(0.528, 0.644, 0.772)	0.6481
$\mathcal{M}_5$	(0.605, 0.722, 0.776)	0.7010

Step 4 We use the known criterion weights for the TFNN hidden layer information in order to proceed. The weights assigned to each criterion in the hidden layer are 0.26, 0.22, 0.31, and 0.21, respectively, indicating their respective significance or impact. Every one of these weights represents the importance of a certain criterion or input characteristic in the TFNN, impacting the network's total output or decision-making. The network modifies its computations to take into consideration the varying weights assigned to each criterion as input moves through the hidden layer by applying these known weight values.

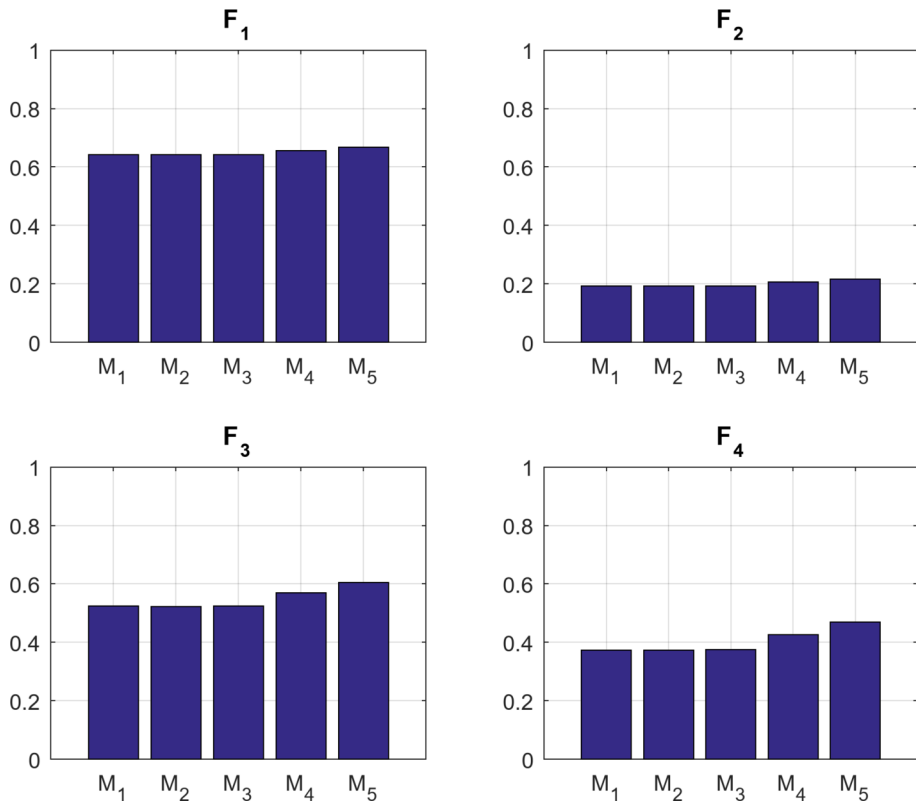
Step 5 Using the Einstein aggregation operator (11), we combine the hidden layer data with their weights to calculate the output layer information. This operator takes into account the importance of each weight when merging the hidden layer data, ensuring an accurate and reliable final output.

Step 6 We use the score function of TFN (12) to compute the score values of the output layer information. Table 4 presents the findings.

Step 7 Finally, we apply the sigmoid activation function, Softmax activation function, hyperbolic tangent activation function, and Swish activation function, as described in Eqs. (13)–(16), to calculate the output of the TFNN and complete the process. These activation functions help convert the unprocessed output of the TFNN into a normalized range, typi-

Table 5 The output of the TFNN

LSPs	$\mathcal{F}_1$	$\mathcal{F}_2$	$\mathcal{F}_3$	$\mathcal{F}_4$
$\mathcal{M}_1$	0.6415	0.1925	0.5239	0.3732
$\mathcal{M}_2$	0.6412	0.1923	0.5232	0.3724
$\mathcal{M}_3$	0.6418	0.1928	0.5250	0.3743
$\mathcal{M}_4$	0.6566	0.2057	0.5704	0.4255
$\mathcal{M}_5$	0.6684	0.2168	0.6050	0.4686

**Fig. 7** Results of the proposed decision making model

cally between 0 and 1, which supports decision making. Based on this, we assess and select the best LSP. Table 5 presents the output results of the TFNN.

The proposed decision making model clearly shows, as seen in Table 5; Fig. 7, that C.H. Robinson Worldwide is the best logistics service provider. The analysis results highlight its better performance compared to other providers, confirming it as the top choice based on the given criteria. Both the data in Table 6 and the graph in Fig. 7 support this conclusion, proving the effectiveness of the proposed model in identifying the best service provider.

6 Comparative analysis

In this section, we evaluate the performance of our proposed decision making model by comparing it with several well-known MCDM methods. This comparison aims to validate the accuracy, reliability, and efficiency of the proposed model in solving complex decision making problems. To conduct this analysis, we use the same expert opinion information matrices related to logistics service providers. These matrices capture the subjective judgments and uncertainties of experts, which are expressed in the form of triangular fuzzy numbers. For this study, the reliability of each expert's information is reflected by pre-defined weights of 0.41, 0.34, and 0.41. These weights play a crucial role in processing the data and ensuring that the most dependable judgments are accentuated in the final decision making process. To aggregate the expert opinions into a single decision matrix, we use the Einstein aggregation operator, as defined in Eq. 8. This operator combines the information provided by individual experts while taking their respective weights into account. The resulting aggregated decision matrix effectively represents the collective judgment of all experts. Next, we incorporate the criteria weights, which are set at 0.26, 0.22, 0.31, and 0.21. These weights indicate the relative importance of each criterion and are essential for conducting the MCDM analysis. By combining the aggregated decision matrix with the criteria weights, we prepare the data for further evaluation. For validation, we compare the outcomes of our proposed decision making model with the results obtained from six well-established MCDM methods: TOPSIS, GRA, WASPAS, MOORA, SWA, and EDAS. Each of these methods is applied to the same decision making problem using the aggregated data. Table 7 provides a detailed comparison of the outputs generated by the different methods, enabling a thorough evaluation of their respective performances. Finally, the results produced by our proposed model are compared with those from the other MCDM methods. This comparison ensures a comprehensive analysis of the model's efficiency and accuracy. The findings, presented in Table 6, demonstrate the validity of the proposed decision making model and highlight its practical effectiveness in decision making scenarios. By aligning well with established methods, our model is shown to be both effective and reliable.

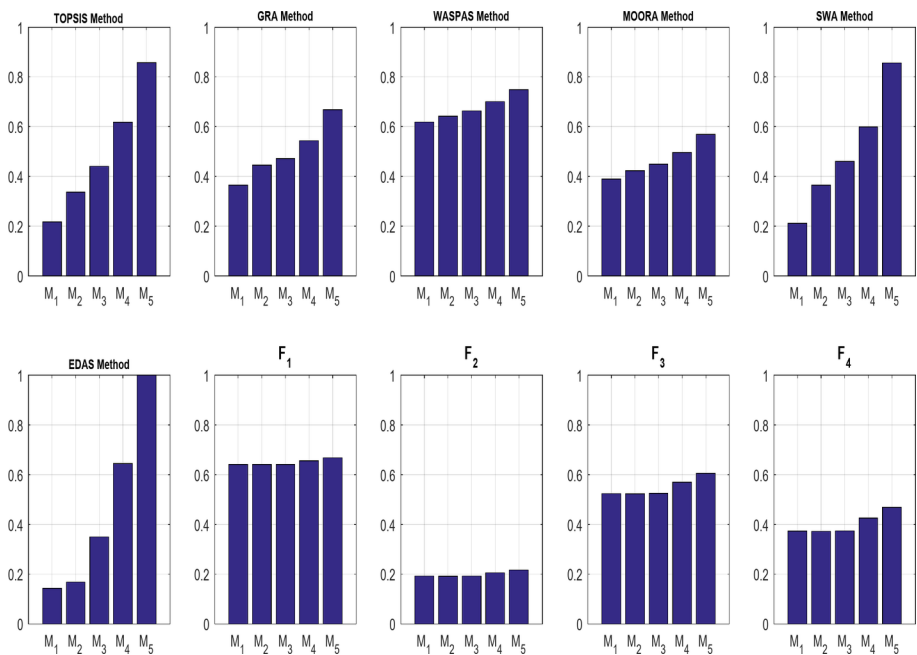
The ranking results show that the outcomes from our proposed decision making model are very similar to those from other well-known MCDM methods. This shows that our model is valid and works well in solving decision making problems. Table 7 presents a detailed comparison of the rankings from our proposed model and the other MCDM methods. It shows that the results are almost the same, proving that our model is reliable and gives consistent outcomes. Figure 8 provides a clear visual comparison of the rankings, showing that C.H. Robinson Worldwide is ranked the highest among the logistics service providers. This confirms that, based on the given criteria, C.H. Robinson Worldwide is the best choice. The comparative analysis demonstrates that our model performs effectively,

Table 6 Results of the comparative analysis based on MCDM methods

Methods	$\widehat{\mathcal{M}}_1$	$\widehat{\mathcal{M}}_2$	$\widehat{\mathcal{M}}_3$	$\widehat{\mathcal{M}}_4$	$\widehat{\mathcal{M}}_5$
TOPSIS	0.2171	0.3359	0.4407	0.6181	0.8569
GRA	0.3645	0.4451	0.4718	0.5433	0.6677
WASPAS	0.6184	0.6416	0.6635	0.7007	0.7490
MOORA	0.3889	0.4228	0.4495	0.4963	0.5683
SWA	0.2107	0.3644	0.4606	0.5986	0.8561
EDAS	0.1432	0.1684	0.3499	0.6450	1.0000

Table 7 Ranking of the output results

Methods	Ranking	Best
TOPSIS	$\widehat{\mathcal{M}}_5 > \widehat{\mathcal{M}}_4 > \widehat{\mathcal{M}}_3 > \widehat{\mathcal{M}}_2 > \widehat{\mathcal{M}}_1$	$\widehat{\mathcal{M}}_5$
GRA	$\widehat{\mathcal{M}}_5 > \widehat{\mathcal{M}}_4 > \widehat{\mathcal{M}}_3 > \widehat{\mathcal{M}}_2 > \widehat{\mathcal{M}}_1$	$\widehat{\mathcal{M}}_5$
WASPAS	$\widehat{\mathcal{M}}_5 > \widehat{\mathcal{M}}_4 > \widehat{\mathcal{M}}_3 > \widehat{\mathcal{M}}_2 > \widehat{\mathcal{M}}_1$	$\widehat{\mathcal{M}}_5$
MOORA	$\widehat{\mathcal{M}}_5 > \widehat{\mathcal{M}}_4 > \widehat{\mathcal{M}}_3 > \widehat{\mathcal{M}}_2 > \widehat{\mathcal{M}}_1$	$\widehat{\mathcal{M}}_5$
SWA	$\widehat{\mathcal{M}}_5 > \widehat{\mathcal{M}}_4 > \widehat{\mathcal{M}}_3 > \widehat{\mathcal{M}}_2 > \widehat{\mathcal{M}}_1$	$\widehat{\mathcal{M}}_5$
EDAS	$\widehat{\mathcal{M}}_5 > \widehat{\mathcal{M}}_4 > \widehat{\mathcal{M}}_3 > \widehat{\mathcal{M}}_2 > \widehat{\mathcal{M}}_1$	$\widehat{\mathcal{M}}_5$
$\mathcal{F}_1$	$\widehat{\mathcal{M}}_5 > \widehat{\mathcal{M}}_4 > \widehat{\mathcal{M}}_3 > \widehat{\mathcal{M}}_1 > \widehat{\mathcal{M}}_2$	$\widehat{\mathcal{M}}_5$
$\mathcal{F}_2$	$\widehat{\mathcal{M}}_5 > \widehat{\mathcal{M}}_4 > \widehat{\mathcal{M}}_3 > \widehat{\mathcal{M}}_1 > \widehat{\mathcal{M}}_2$	$\widehat{\mathcal{M}}_5$
$\mathcal{F}_3$	$\widehat{\mathcal{M}}_5 > \widehat{\mathcal{M}}_4 > \widehat{\mathcal{M}}_3 > \widehat{\mathcal{M}}_1 > \widehat{\mathcal{M}}_2$	$\widehat{\mathcal{M}}_5$
$\mathcal{F}_4$	$\widehat{\mathcal{M}}_5 > \widehat{\mathcal{M}}_4 > \widehat{\mathcal{M}}_3 > \widehat{\mathcal{M}}_1 > \widehat{\mathcal{M}}_2$	$\widehat{\mathcal{M}}_5$

**Fig. 8** Graphical representation of the comparative analysis

delivering reliable and consistent results for decision making, particularly in the logistics sector.

7 Advantages and limitations of the proposed approach

In this section, we discuss the advantages and limitations of our proposed triangular fuzzy neural model for decision making problems, particularly focusing on its contributions to addressing complex and uncertain environments.

Advantages of the proposed approach:

- i. **Flexibility across applications:** The proposed TFNN model is adaptable to various MCDM problems, including healthcare, industrial material selection, and e-commerce. For example in logistics, the model evaluates service providers by incorporating multiple uncertain criteria such as cost, timeliness, and reliability.
- ii. **Enhanced handling of uncertain data:** By using triangular fuzzy numbers, the model captures and processes imprecise input data more effectively than traditional methods. For example in selecting nanomaterials, TFNs represent uncertain factors like durability, thermal resistance, and cost.
- iii. **Improved decision making accuracy:** The integration of the Einstein aggregation operator enhances the ability to combine multiple criteria while maintaining consistency. For example this ensures more accurate rankings of logistics service providers, even when criteria weights exhibit uncertainty.
- iv. **Scalability for big data applications:** The TFNN model can handle large datasets efficiently, making it applicable to modern big data environments. For example in e-commerce, it predicts customer preferences from large, fuzzy datasets, enabling better product recommendations.
- v. **Robust aggregation tools:** The Einstein aggregation operator, tailored for triangular fuzzy numbers, ensures smooth handling of multi criteria input while addressing inter-dependencies between criteria.
- vi. **Practical applicability:** The model demonstrates applicability to real-world decision making tasks, such as logistics provider evaluation, by considering ambiguous and uncertain input data.

Limitations of the proposed approach:

- i. **Dependence on complete input data:** The model requires complete and accurate input data for optimal performance. For example the missing or incomplete data can hinder its effectiveness, such as incomplete environmental data in urban planning applications.
- ii. **High computational complexity:** The computational requirements of the TFNN model can be high, especially when dealing with large-scale problems or extensive criteria. For example the evaluating a large number of alternatives with multiple criteria in aerospace engineering demands significant computational resources.
- iii. **Sensitivity to weight calculation methods:** The model's effectiveness relies on the accurate calculation of criteria and hidden layer weights.

Advantages of triangular fuzzy numbers in neural networks.

- i. **Representation of uncertainty:** TFNs effectively model imprecise and ambiguous information, making them ideal for real-world decision making scenarios where

- uncertainty is essential. For example when evaluating logistics providers, TFNs capture subjective assessments like service quality or customer satisfaction.
- ii. **Simple yet Effective:** TFNs offer a balance between computational efficiency and the ability to represent uncertainty, compared to more complex fuzzy numbers like trapezoidal or interval type-2 fuzzy numbers. For example this simplicity ensures faster processing without sacrificing decision-making accuracy.
 - iii. **Facilitates smooth integration:** TFNs integrate easily with aggregation operators like the Einstein operator, enhancing the robustness of neural network models.
 - iv. **Improved interpretability:** The triangular structure of TFNs (lower, middle, and upper bounds) is intuitive and easier to interpret, aiding decision-makers in understanding the results.
 - v. **Wide applicability:** TFNs are useful and can represent diverse types of uncertain criteria across industries, from healthcare to logistics.

The proposed triangular fuzzy neural network model offers significant advantages in handling complex and uncertain decision making scenarios. While it has certain limitations, such as computational complexity and reliance on complete data, these are mitigated through efficient aggregation techniques and robust weight calculation methods. The use of triangular fuzzy numbers further enhances the model's capability to represent and process imprecise information, making it a valuable tool for modern decision making challenges. Future work will focus on addressing these limitations by developing lightweight versions of the model and exploring its application in broader contexts.

8 Conclusions

In this article, we introduced a novel fuzzy neural network approach based on triangular fuzzy numbers to solve MCDM problems. The goal was to help choose the best logistics service providers, which is a critical task for large e-commerce businesses. Logistics service providers help with the cost-effective processing, storing, and delivery of products, but deciding which one is the best is not easy because many good providers are available. The proposed decision making model uses a triangular fuzzy neural network combined with a triangular fuzzy Einstein weighted averaging operator to solve this problem. We first explained the concept of triangular fuzzy numbers, including their score function and Hamming distance measure. These tools help us handle uncertainty in the decision making process. Then, we defined the operational laws of TFNs and the TFEWA operator based on Einstein norms. The TFEWA operator is important because it helps combine expert opinions in a way that makes the decision making process more reliable. We then described how the TFNN works in the context of MCDM problems. Expert opinions about the logistics providers were collected and used as input information for the TFNN. The TFEWA operator was then used to combine the input information with corresponding weights. We applied the TFEWA operator once more to get the output information, and then used a score function to find the score values in the output layer. These values were processed using an activation functions to produce the final result. The outputs create a ranked list of logistics service providers based on their performance in different areas. The decision-making model clearly shows that C.H. Robinson Worldwide is the best provider. The analysis reveals that it performs better than other providers, making it the top choice according to the criteria used. This confirms that

C.H. Robinson Worldwide stands out as the best option among all the providers, based on the evaluation of their overall performance. Finally, we compared our method with other common MCDM techniques to check its effectiveness. The results showed that our approach is efficient, reliable, and better at handling uncertain data. Overall, the proposed method provides a powerful tool for selecting the best logistics service provider.

In future research, we aim to address the high computational complexity by optimizing the triangular fuzzy neural network model for large-scale problems. We plan to explore lightweight versions of the model or more efficient algorithms that can reduce computational requirements without sacrificing accuracy. Additionally, we will work on improving the weight calculation methods by investigating alternative techniques that may offer better accuracy and robustness. We will also extend the Aczel-Alsina aggregation for triangular fuzzy numbers to handle more complex decision-making scenarios, which will further enhance the flexibility and applicability of the model.

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Data availability No datasets were generated or analysed during the current study.

Declarations

Competing interests The authors declare no competing interests.

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